



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-term Exam, Trimester: Fall 2025

Course Code: CSE 3521, Course Title: Database Management Systems

Total Marks: 30 Duration: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. a) You are tasked to design an e-commerce platform. The platform hosts multiple vendors, each having a unique vendor ID. Each vendor has name, contact numbers (multiple possible) and address information which consists of street, city and country. Each vendor can sell multiple products. Each product will contain a unique product ID, name, category, price, and stock quantity information. 7

The platform serves the customers that are identified by customer ID. Customers have other attributes including name, email, phone numbers (multiple allowed), and shipping addresses that consists of street, city and country information. Customers can place multiple orders that are identified by order ID. Each order contains information about the payment method, discount amount, total amount, order date, and delivery status. An order can belong to only one person.

Each order contains order items that can be identified by the order id and the product id of that item. Quantity and subtotal for each order item is also recorded. Customers may write reviews for products that contain a unique review ID, rating, review date, and comments. Vendors may collaborate with other vendors in partnership agreements.

Now, **design** an ERD for the system, clearly outlining its entities, attributes, and relationships with appropriate cardinality.

b) What is discriminator? **Explain** the role of identifying entity in discriminator. 2

2. **Draw** the corresponding schema diagram based on the erd given in Figure 1. 6

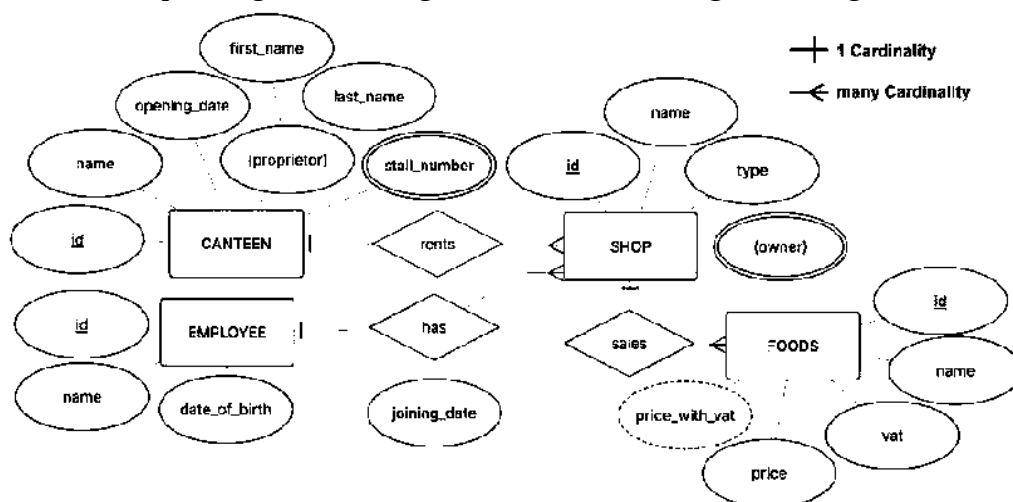


Figure 1

3. Consider the following relational database schema:

Users(user_id, username, email, location, signup_date)

Posts(post_id, user_id (fk), content, timestamp, likes_count)

Products(product_id, name, category, price, stock_quantity)

Orders(order_id, user_id (fk), order_date, total_amount)

OrderDetails(order_id (fk), product_id (fk), quantity, price)

Recommendations(rec_id, user_id(fk), product_id(fk), algorithm, score)

Reviews(Review_Id, review)

Answer the following questions using **relational algebra**:

- a) List all products in the category 'Electronics' with their name and price. 1
- b) Retrieve the username and product name for all products a user has ordered. 1
- c) Write a sentence describing the output of the following relational algebra: 2
$$\sigma_{\text{category}='Electronics'}(\text{Products} \bowtie \text{OrderDetails}) \cap \sigma_{\text{category}='Electronics'}(\text{Products} \bowtie \text{Recommendations})$$

4. Using the schema given in Q3, **write sql queries** for the below.

- a) Add a new column user_id in Review table and ensure that it references the appropriate table. 2
- b) Increase price of products whose price is below \$100 by 5%, and all others by a 2%. 2
- c) Find the name of the product(s) which has the highest recommendation. 2
- d) Find the user list with their total order amount whose name ends with R and atleast 4 characters long. 2
- e) Find all users ID who have placed orders containing products from the 'Electronics' category, where the *total quantity ordered* by each user for the 'Electronics' products is greater than 5. 3